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Distributional properties of English near synonyms in lexical domains: a formal and metric-based approach

Abstract

The present study first proposes a formal account for identifying sets of near synonyms from the same lexical domain through the reduction of polysemy. Secondly, it introduces metrics for investigating the distributional properties of near synonyms from the same lexical domain. The focus of the present study is on 'verbs of destruction' and all the data are obtained from the British National Corpus using the Sketch Engine tools. Combining insights from theoretical linguistics, corpus linguistics and distributional semantics, the study highlights the importance of both collocational relations and syntactic and thematic structures in understanding the lexical meaning of 'verbs of destruction'. The formal approach models polysemy reduction as an incremental process in which contextual elements disambiguate meanings and establish clear criteria for near synonymy within a specific lexical domain. A lexical domain is defined as a set of synonymous lexical items that share a common collocate and occur in a specific syntactic and thematic structure. Furthermore, the study introduces novel metrics, Total Strength, Breadth and Average Strength and Typicality Index, which quantify the representativeness, breadth of usage and association strength of a verb within a lexical domain. The metrics allow the investigation of different aspects of verb behaviour within the lexical domain, such as the cumulative effect of association strength, generality vs. specificity, and broad applicability vs. niche use.

Keywords

*near synonymy, lexical domains, polysemy reduction, metrics, syntactic and thematic structures
distributional semantics*

Introduction

This paper has two primary aims. First, it proposes a formal account of the procedure for identifying sets of near synonyms through the reduction of polysemy. Second, it introduces metrics for examining the distributional properties of near synonyms. Both the grouping of lexical items into synonym sets and the semantic differentiation between them have been explored in traditional lexicology (e.g., Trier, 1939; Lehrer, 1974; Cruse, 1986; Van Roey, 1990; Miller and Fellbaum, 1991) and in corpus-oriented lexicology (e.g., Halliday et al., 2004; Teubert, 2005; Divjak, 2010; Glynn, 2014). At first glance, it may seem paradoxical to bring

these two processes together, since they are based on two mutually exclusive assumptions. The identification of synonymy sets relies on the assumption that the lexicon contains items that are semantically identical or very similar. On the other hand, the differentiation of meaning between lexical items is based on the assumption that lexical items differ in meaning. However, when seen through the lens of linguistic gradience, polysemy and synonymy reflect two sides of the same phenomenon: the dynamic interplay between similarity and distinction in language. As is well-known, full synonymy is very rare in language (e.g., Cruse, 1986). Grouping words into synonym sets operates at a macro-semantic level, focusing on broad overlaps in meaning. Differentiation, by contrast, functions at a micro-semantic level, where subtle distinctions between near synonyms are analysed (for a corpus linguistic approach see for example Gries and Divjak, 2009; Gries, 2010; Gries and Otani, 2010). This duality reflects the way language balances redundancy (grouping similar meanings) with precision (marking subtle distinctions). The complementary nature of polysemy and synonymy can be graphically explained in terms of the meaning/form distinction. In polysemy, a single linguistic form is associated with several meanings that are semantically related (one form, more than one meaning) (Gries, 2019). For example, the noun *run* has 16 different senses listed in WordNet, including the sense ‘an unbroken chronological sequence’, as in *The play had a long run on Broadway*, or ‘a short trip’, as in *take a run into town*. Sometimes, the meaning depends on the part of speech. Thus, none of the 41 senses of the verb *run* in WordNet includes the sense of ‘an unbroken chronological sequence’ associated with the noun *run*. Additionally, the specific sense may depend on the syntactic structure of the lexical item. For example, the verb *run* differs in meaning when used intransitively (*Mary runs every morning*) versus transitively (*Mary runs a company*). In synonymy, different linguistic forms share the same or rather very similar meanings (more than one form, one meaning). For example, *begin* and *start* both denote the initiation of an event. However, full synonymy is rare in practice, and these words may differ in terms of register, nuances of meaning, or the collocates they attract. In my previous study (Trklja, 2014), I demonstrated that even highly contextualized terms that function as synonyms and as translation equivalents in another language may differ in terms of the additional collocates they can combine with. Verbs can also differ in terms of their argument structure, being synonymous in some syntactic frames but not in others (see Section 2 below). In studying lexical relations, the focus is on the lexical item’s specific meanings realized in a specific context rather than on all its possible meanings. Therefore, whenever a lexical item is polysemous, the identification of synonyms involves reducing the polysemy of a term to its contextually relevant senses, emphasising its overlap with other words in the set.

To achieve the aims from above, the present study explores synonymy relations and meaning differentiation in terms of assumptions derived from distributional semantics enhanced by insights from theoretical linguistics. In particular, the identification of synonymous relations will be based on collocational relations. Additionally, as discussed in the next section, I will introduce the requirement that collocates must meet specific criteria regarding their position in the syntactic and semantic structure. While the methodological framework is adaptable to any lexical domain in any language, the analysis is limited here to English lexical domain of ‘verbs of destruction’ for illustrative purposes. Data were obtained from the British National Corpus (BNC) (Leech, 1992) using the Sketch Engine tools (Kilgarriff et al., 2014). Again, as the purpose of the paper is not to explore a particular domain of lexical meaning or to test specific corpus tools, any other corpus with sufficient data and any other tool capable of the relevant functionality could be used.

Exploring lexical semantics of near synonymy

According to the traditional lexical field theory “the vocabulary of a language is structured, just as the grammar and phonology are structured” (Lehrer, 1974: 15). In particular, it was argued that the vocabulary or the lexicon (as we would more commonly say nowadays) is organised into the classes of synonymous terms. Until recently, most approaches to lexical semantics in general and to the study of semantic similarity (e.g., Jackendoff 1988, 1992; Levin 1993; Talmy, 2003) have focused on the conceptual nature of meaning, reflecting the ‘turn to conceptuality in the 1980s (Lang and Maienborn, 2011). These approaches examined the meaning of lexical items (primarily single words) in terms of theoretical constructs such as semantic primitives, viewing semantic structure as a type of conceptual structure. This view is rooted in Katz and Fodor’s (1963) argument that meaning can be decomposed into a finite set of semantic primitives. The semantic primitive approach has been widely used to investigate how lexical items are grouped into semantic classes, as well. Lehrer who argued that the lexicon of a language consists of word sets “which are related to conceptual fields and [that] divide up semantic space or the semantic domain in certain ways” (Lehrer, 1974: 15) identified a range of semantic components that characterise the meaning of words in the semantic field COOKING (e.g. ‘the use of water’, ‘the use of oil’, ‘the cooking time’ and ‘the use of cooking utensils’). The major problem with this and similar studies from lexical field theory is the lack of a generally accepted method for identifying such components (Trklja, 2017). While the literature often assumes that semantic primitives are universal, critics have noted (Gordon,

2003) the lack of universal principles for their identification. Another challenge for this view is the delineation of semantic boundaries among lexical items. As Jackendoff (1983:113) observes, while the theory of decomposition “nicely satisfies the common sense intuition that words have definite and precise meanings” the semantics of lexical items often exhibit fuzzy boundaries. Indeed, as he further notes, “fuzziness is an inescapable feature of the concepts that language expresses” (Jackendoff, 1983:117). Besides, such studies face a circularity problem: the semantic components selected to define the members of a semantic field are often identified only after the fields themselves have been established. Consequently, the identification of the synonymy sets relies more on the researcher’s intuition and world knowledge than on the intrinsic linguistic properties of lexical items.

The notion of semantic primitives has also found its way into more recent corpus-based approaches to semantic fields (e.g. Viberg, 2002, 2008; Boas, 2002; Dyvik, 2004), which have attempted to overcome the drawbacks of the traditional lexical field approach by using corpus linguistic techniques and insights from cognitive linguistics. Viberg (2002, 2008) thus suggests that there is a set of basic meanings that are universal across languages, around which semantic fields are organised. According to the author, the basic meanings are represented by one of the 20 most common verbs in different languages, including BE, CAN, GIVE, TAKE, SAY, SEE. These primitives can apparently then be used to disambiguate the polysemy of lexical items and group them into semantic sets. For example, the Swedish verb *få* has different meanings, and when it denotes possession, it can be described by the semantic primitive GET. But here the problem from above remains since there are no universal principles that should ensure that exactly these semantic primitives exist and not others. In fact, if one compares the list of semantic primitives proposed by Viberg with that of Natural Semantic Metalanguage (NSM) (Wierzbicka, 1996), one can notice that despite some overlap the two lists differ significantly. They differ because they are based on different assumptions and it is just not clear which assumption should be trusted. Another problem is that the semantic primitives are referred to here in terms of actual English verbs which are themselves polysemous, and it is not obvious which sense is meant when they are used as names for semantic primitives. In fact, in order to define their senses, we would have to assume another set of semantic primitives, which would again be represented in terms of words that have different senses, and so we enter the endless regress of definitions. It is here worth mentioning Boas's (2002, 2005) approach to semantic grouping, which builds on the concept of semantic frames from frame semantics (Fillmore, 1976; Fillmore and Atkins, 1992), also assumed to represent a universal category. In exploring

the semantic frames of German verbs, he starts from the assumption that a frame serves to as “an independently existing conceptual system that is not bound to any particular language” (Boas, 2005:466). However, the frames he applies to German expressions are not really universal, since they are derived from frames identified in the study of English expressions. As a result, only phenomena observed in English can be reflected in the German data and any German-specific features that are not present in English will not be captured by such an approach. Hanks (2004:6) criticises, the frame semantics approach because it “requires researchers to think up all possible members of a frame a priori, [which] means that important senses of words that have been partially analysed are missing and may remain missing for years to come”. As a result, many frames remain incomplete and many word senses are not covered. The core problem is that this approach starts with predefined frames rather than empirical data. As Hanks (2004:6) points out “[w]hat is needed is a principled fix - a decision to start with evidence, not frames. This is precluded by FrameNet (Fillmore and Johnson, 2003; Baker, 2012) on principled grounds: the unit of analysis for FrameNet is the frame, not the word”. Moreover, only formally similar frames are matched across languages, whereas the same meaning is often expressed by different syntactic frames.

Recently, distributional semantics has emerged as a novel approach that can be used to identify synonymy sets. The approach has its origins in the structural linguistics of Zellig Harris, and the following much-quoted quote captures the distributional perspective on meaning:

“If we consider words or morphemes A and B to be more different in meaning than A and C, then we will often find that the distributions of A and B are more different than the distributions of A and C. In other words, difference in meaning correlates with difference in distribution” (Harris, 1970:785).

In this view, the distribution of a linguistic element is defined as the set of textual environments in which it occurs (Harris, 1952, 1954). Harris (1952) further posits that lexical items that share contextual occurrences form equivalence classes are mutually substitutable. This notion of substitutability echoes Leibniz’s definition of identity: “Two things are the same if one can be substituted for the other without affecting the truth” (Lyons, 1977:160). Harris’s focus was on morphemes and words but in the more recent computational linguistics approaches the distributional hypothesis has been applied to the study of larger linguistic units, including multi-word units. These new approaches make use of corpus data to develop computational

distributional semantics models for representing meaning (Lenci and Sahlgren, 2023). These models assess similarity relations by measuring the typicality of co-occurrence patterns using statistical association measures (Manning and Schütze, 1999; Evert, 2005). The resulting co-occurrence matrices are reduced in dimensionality using techniques such as singular value decomposition or truncated singular value decomposition. In this way, the essential patterns are identified and the semantic relationships between linguistic items are represented as vector spaces. Proximity in vector space, often measured by cosine similarity, indicates semantic similarity, with words whose vectors are closest being considered most similar. While distributional approaches can identify subtle distributional differences and contextual nuances, they often lack explanatory power for lexical semantic analysis. As Lenci and Sahlgren (2023) note, these methods reduce semantic analysis to identifying syntactic and lexical collocates, classifying lexemes into classes, and relying heavily on co-occurrence relations. This is because the purely collocational approach following Wittgenstein (1953) equates meaning with usage. Nevertheless, there are important aspects of lexical meaning that are not fully captured by the distributional view such as the lexical aspect, thematic relations or pragmatic meaning. The same problem is observed in the Sinclairian corpus-driven approach to meaning (e.g. Tognini-Bonelli, 2001), which shares the meaning-as-use assumption and operates under a minimal assumption approach, according to which at the outset “[w]e should only apply loose and flexible frameworks until we see what the preliminary results are in order to accommodate the new information that will come from the text” (Sinclair, 1994: 25). This is a word-centred view of the relationships between lexical items in a sentence, but the major syntactic theories agree that sentences are not made up of words but of syntactic phrases which are arranged hierarchically and which make up the syntactic structure of the sentence. The pure co-occurrence approach cannot capture such hierarchical relations. This is well illustrated by Divjak’s (2010) study of near synonyms in the Russian lexicon, which includes syntactic information in the analysis in addition to collocational relations. In particular, Divjak integrates a corpus-based distributional approach to meaning with a construction grammar perspective. According to the CxG view, syntactic structures are not autonomous but deeply intertwined with semantics. Divjak proposes “that the closer any two verbs are in their semantic structure, the greater the overlap should be in their licensed syntactic structures” (Divjak, 2010:12). I agree with the assumption that syntactic relations and especially argument structure must be included in the study of lexical meaning in general and near synonymy in particular but I consider this particular assumption too strong. Let us look at this in a little more detail.

As Levin's (1993) work on alternation patterns show verbs may be mutually near synonymous although they do not participate in the same alternation patterns. Consider the verbs *fill* and *pour* both of which can collocate with *water*.

1. Leslie filled the bottle with water.
2. *Leslie filled water in(to) the bottle.
3. The water filled the bottle.
4. The bottle filled with water.
5. *Leslie filled the bottle in water.

6. *Leslie poured the bottle with water.
7. Leslie poured water into the bottle.
8. *The water poured the bottle.
9. *The bottle poured with water.
10. *Leslie poured the bottle in water.

The difference between these synonymous verbs lies in how they assign thematic roles and their focus within argument structures. In [1] and [4] the Goal (e.g., *the bottle*) is used as the direct object, emphasizing the resulting state of the container, while the Theme (e.g., *water*) is introduced via a prepositional phrase (*with water*). This structure supports alternations where the Goal can function as the subject (*The bottle filled with water*) or the object (*Leslie filled the bottle with water*), and even allows the Theme to act as the subject (*The water filled the bottle*). In contrast, *pour* focuses on the transfer of the Theme (substance), which is always the direct object (*Leslie poured water into the bottle*), while the Goal is introduced via a preposition (*into the bottle*). The Goal cannot take the role of subject or object, and no alternations like those seen with *fill* are possible. Of five possible alternation patterns, *fill* accommodates three whereas *pour* allows only one highlighting its stricter focus on the movement of the Theme. This distinction underscores how *fill* emphasizes the state of the Goal, while *pour* prioritises the action involving the Theme.

Similarly, as demonstrated in my previous research (Trklja, 2020), verbs that share the same collocates can occur with different syntactic and semantic structures while still being near synonyms. Consider the examples [11-14].

11. The inflation caused problems.

12. Inflation led to problems.
13. Inflation gave rise to problems.
14. Problems arose due to inflation.

In these examples, the verbal predicates (*cause*, *lead to*, *give rise to*, *arise*) all describe a causal relationship between *inflation* and *problems*, but their syntactic structures and thematic roles differ. *cause* is a transitive verb that assigns the Agent to the nominal phrase in the subject position that denotes a causer to *inflation* and a Theme to *problems*. *lead to* introduces result (*problems*) via a prepositional phrase, focusing on the process of causation. *give rise to* employs a metaphorical ditransitive construction where inflation is the source and problems the result. Strictly speaking, *rise* is a Theme and problems is a Beneficiary. Finally, *arise* is an intransitive verb where *problems* serves as the Theme, and the cause (*inflation*) is introduced as an Instrument through a prepositional phrase (*due to inflation*). These differences illustrate how near-synonyms vary in argument structure and the thematic role assignment while expressing overlapping meanings.

Finally, some verbs might share the same syntactic and semantic structures, even though they are not synonymous as in [15] and [16].

15. Break: She broke the vase / The vase broke.
16. Open: She opened the door / The door opened.

Although Divjak's assumption is too strong, it highlights the importance of considering information about syntactic and semantic structure in the exploration of lexical meaning and synonymy, which is, as we said, neglected in corpus linguistic and distributional semantic approaches. I will take up this point, albeit in a different way. Following the dominant view in syntactic theory (e.g., Koenenman and Zeijlstra, 2017, and for a more theoretical elaboration, see Williams, 1994; 2002) I will consider that each verb is associated with one or more argument structures and that these structures receive semantic interpretations, typically referred to in the literature as thematic structures. Verbal predicates impose syntactic and semantic constraints on how their arguments combine with them, ensuring alignment between syntax and meaning. Each verb has one or more inherent argument and thematic structures that specifies the roles of its arguments, such as *READER* (Agent) and *READ* (Theme) for the verb *read*. These thematic roles act as semantic requirements for the syntactic constituents that accompany the verb. In the present example, the subject must align with the role of the Agent (e.g., someone capable of reading), and the object must align with the role of the Theme (e.g.,

something that can be read). This relationship ensures that syntactic categories (like subjects and objects) are not combined arbitrarily but must semantically fit the verb. For instance, in the sentence *Mary reads a book*, Mary satisfies the READER role, and a book satisfies the READ role. Violating these constraints, such as in *A book reads Mary* results in semantic incongruity. Thus, thematic roles serve as a bridge between syntax and semantics, restricting the combination of constituents with verbs to maintain coherence and meaning.

For the reasons just explained, in order to overcome the aforementioned limitations of the distributional approach, I propose that it is necessary to take into account both syntactic and semantic relations between collocates. Specifically, since the present study is concerned with verbs and their collocates the focus will be on verbal phrases (VP). In particular, a node word selected for analysis will be a head verb and its collocate(s) will be the syntactic argument(s) of the head verb. Due to the argument structure restriction, it will be required that words collocating with a verb are part of the same syntactic constituent. This means that verbs will be considered near synonyms and the candidates for the same lexical domain if they collocate with the same word that also has the same constituent function (e.g. a direct object). It is important to note that a constituent may be realised differently, but in all cases the collocating word must be linked to the head verb in the same way. For example, in *create problems* and *lead to problems*, the noun *problems* is part of an internal argument of the head verb, realised as a part of a noun phrase in the first case and as a part of a prepositional phrase in the second. Such a restriction also covers cases where verbs associated with different syntactic alternations still belong to the same lexical domain. However, it does not take into account cases where a collocate occurs in different syntactic positions. For example, *problems* acts as a direct object with *cause* and as a subject with *arise*. But, despite these syntactic differences, both collocations can be considered part of the same lexical domain, as explored in my previous research (Trklja, 2014, 2020). Crucially, although the arguments of the verb are syntactically realised differently, the collocating nouns are assigned the same semantic role (i.e. Theme).

Therefore, the argument structure constraint must include both syntactic and thematic relations. I will therefore propose that verbal expressions belong to the same lexical domain if the head verbs share the same collocate and if they share either the same argument or semantic structure, or ideally both. This dual criterion ensures that the classification reflects both structural and semantic coherence, thus overcoming the problem of Divjak's strong assumption.

Incremental reduction of polysemy and lexical domains

As was mentioned above, polysemy and synonymy are two interrelated phenomena. When exploring near synonymy, it is essential to define the sense(s) in which the terms are semantically similar. This process inherently involves disambiguating the meanings of the terms to identify the specific sense(s) in which they overlap. Synonymous terms cannot be established by examining words in the totality of their meanings, because synonymy is not a property of words in isolation but rather a function of how terms are used in specific collocational contexts. *brilliant* and *bright* are not simply synonymous in all their senses since *brilliant* overlaps with *bright* in the sense of intelligent but not necessarily in the sense of shining. A central tenet of corpus linguistics and distributional semantics is that context specifies lexical meaning. However, implementing and formalizing this idea is not a trivial task as illustrated in Manning and Schütze (1999). I will argue here that the more detailed the context, the more specific the meaning of a word. Consequently, I will propose that contextualisation of meaning should be considered as a reduction of polysemy in terms of degree. The main idea is that by incrementally increasing the number of contextual elements we establish the individual senses of a polysemous term. A clarification is in place here. Additional context might introduce nuances or secondary layers of meaning (e.g., modifiers affecting style or register) that does not typically change a core lexical sense of a word. For example, adverbs that add nuances to certain aspects of meaning such as tone or register but do not disambiguate between core senses would be considered peripheral in the present framework. Suppose that *break* has the following three meanings:

- a) to physically fracture something.
- b) To violate a rule or a promise.
- c) To interrupt silence.

The adverb *deliberately* modifies each of these senses adding the meaning of an intentional action carried out by the agent. However, it does not cause *break* to take on a new sense and it still denotes the same three types of event as when used without this adverb (the events of physically fracturing something, breaking a rule and interrupting silence).

- 17. He deliberately broke the vase during the argument.
- 18. But last night, they deliberately broke their own rules.
- 19. He deliberately broke the silence with a loud cough.

To address the ambiguity surrounding the term ‘contextual element’, I propose the following assumption based on the discussion from the previous section: verbal expressions are near synonyms and belong to the same lexical domain if the head verbs share the same collocate that appears in a defined syntactic position and/or fulfils a specific thematic role in relation to the head verb. This assumption ensures that the contextual elements are grounded in collocational relations and in both syntactic and semantic criteria, reflecting the complex relations between form and meaning in natural language. In summary, verbs sharing the same collocate in the same syntactic position and/or thematic structure are considered near synonyms and belong to the same lexical domain. The crucial requirement is that the collocate in question must disambiguate the sense of the verb, ensuring that the shared domain is based on stable and semantically meaningful senses. Obviously, polysemous verbs belong to more than one lexical domain, since the same term occurs with different collocates and belongs to separate domains based on the disambiguating collocate. In the following, I will provide a formalization of this idea. Let us define the concepts of polysemy and monosemy for a given verb v as follows:

$$M(v) = \{m_1, m_2, \dots, m_n\}, \text{ where } m_i \text{ a distinct meaning of the verb } v$$

$$M(v) > 1: \text{The verb } v \text{ is polysemic.}$$

$$M(v) = 1: \text{The verb } v \text{ is monosemic}$$

Let C represent the set of all possible contextual elements. A specific context c is defined then as a subset of C :

$$c \subseteq C$$

We can now define a reduction function, $R(c, v)$, that maps a verb v and a given context c to a subset of $M(v)$ where:

$$R(c, v) \subseteq M(v).$$

The size of $R(c, v)$ determines the level of ambiguity of v (v -element) in context c (c -element) allowing us to define context-dependence of polysemy and monosemy as:

$$R(c, v) > 1: \text{The verb } v \text{ remains polysemic in } c.$$

$$R(c, v) = 1: \text{The verb } v \text{ is monosemic in } c.$$

Although I follow the assertion by Kilgarriff (1997) and Hanks (2008) that words are inherently polysemous under specific circumstances (e.g. in the case of terminology) it is possible that a single word is semantically fully disambiguated. Therefore, we must add this theoretical possibility by indicating that the context can be also the empty set, which I define as “zero context”. In addition, I introduce here the notion of cardinality to refer to the size of a set (i.e., the number of elements in it). If there are no co-occurring words, the set c is an empty set. The reduction function R is still valid in this case since $R(c, v) \subseteq M(v)$.

If $R(c, v) = 1$ then the verb v is inherently monosemic without co – occurring words. We can also say that the cardinality of co – occurring words is zero

If $R(c, v) > 1$ then the verb v remains polysemic in zero context, requiring additional contextual items for disambiguation.

Finally, we can define degree of specification, $k(v)$, that measures the minimum number of contextual elements required to reduce the meanings of v to monosemy in the following manner:

$$k(v) = \min\{|c| : R(c, v) = \{m_i\}, \text{ where } m_i \in M(v)\}.$$

To distinguish between inherently polysemic and effectively monosemic verbs, we define a threshold T :

If $k(v) > T$ then the verb v is considered inherently polysemic and resistant to easy disambiguation.

If $k(v) \leq T$ then the verb v is effectively monosemic under specific circumstances.

Here the value of T must be determined empirically where strategy involves starting from zero context (no co-occurring items) and incrementally increasing the cardinality of the contextual set analysing how additional context affects the resolution of ambiguity. At each step k , assess whether $R(c, v) = 1$, indicating that the item becomes monosemic and its meaning resolves completely within the given context. We continue adding contextual elements until semantic ambiguity persists and a specific sense is being determined. For each k , the reduction function $R(c, v)$ operates in order to narrow down the possible meanings of the lexical item v . The

stabilization point corresponds to the degree of specification $k(v)$ or the minimum cardinality of c required to identify a lexical domain. The process is inherently adaptive: for some items, k might stabilize at a zero or low value, while others might require more context. To illustrate how this works suppose that the verb *run* has the following three senses represented using quotation marks:

$$M(\emptyset, run) = \{ 'move quickly', 'operate', 'manage' \}.$$

It can be observed that at zero context, *run* is highly polysemic ($R(\emptyset, run)=3$) and that disambiguation is not possible. If we now add two new c-elements to its context ($k=2$) using the reduction function a specific sense of the verb is being established which can be represented as $R(\{a\ company\}, run)=\{ 'manage' \}$. Further observation indicates that adding more elements (e.g. *effectively*) specify additionally the meaning of manage but it does not further disambiguate new specific senses of *run*. Therefore, we can conclude that for the sense ‘manage’ for this verb has stabilisation point of two ($k("run")=2$).

This approach is directly applicable to the identification and delimitation of lexical domains. The stabilisation point $k(v)$, where semantic ambiguity is resolved, signals the boundary of a lexical domain. For example, in the case of the verb *run*, the addition of contextual elements such as a *company* reduced its polysemy by specifying the sense “manage”. If some other verbs occur with the same sense they will be considered near synonyms and therefore as the members of the same lexical domain. The process is inherently adaptive: different lexical items might require different levels of contextual detail to stabilise their meanings. By systematically increasing the number of contextual elements, the reduction function $R(c, v)$ narrows down the possible meanings, allowing the precise identification of lexical domains and their boundaries.

Formally, a lexical domain $\mathcal{L} = \{ v \mid \exists c \subseteq C \text{ such that } R(c, v) = \{ m_i \} \wedge R(c, v') = \{ m_i' \} \wedge m_i \sim m_i' \wedge \forall v' \in \mathcal{L} \}$.

Here

- $\exists c \subseteq C \text{ such that}$ states that there exists a context c (a subset of possible contexts) satisfying the following conditions
- $R(c, v) = \{ m_i \}$ indicates that the collocate c disambiguates v to some specific sense m_i
- $R(c, v') = \{ m_i' \}$ indicates that another verb v' in the same context disambiguates to m_i'
- $m_i \sim m_i'$ indicates that m_i and m_i' are near-synonymous.

- $\forall v' \in \mathcal{L}$ indicates that these conditions hold for all v' in the given lexical domain.

For example, the verbs *cause*, *trigger* and *provoke* form the same lexical domain $\mathcal{L} = \{\textit{cause}, \textit{trigger}, \textit{provok}\}$ if the collocate $c = \{\textit{anexplosion}\}$ occurs in the same syntactic position and disambiguates all three verbs to the sense ‘initiate’. $R(\{\textit{anexplosion}\}, \textit{cause}) \ R(\{\textit{anexplosion}\}, \textit{trigger}) \ R(\{\textit{anexplosion}\}, \textit{provoke}) = \{\textit{initiate}\}$. Here, the word in the first placeholder position within the curly brackets (in this case, a noun) is a collocate, while the word in the second position (a verb) is the head word being disambiguated. The verbs *cause*, *trigger* and *provoke* are disambiguated here by the same noun forming a common lexical domain. Applying the same formal procedure and formalism to the lexical domain of ‘verbs of destruction’ yields the following:

$$R(\emptyset, \textit{destroy}) = \{\textit{'defeat'}, \textit{'physical destruction'}, \textit{'termination'}\}$$

$$R(\{\textit{building}\}, \textit{destroy}) = \{\textit{'physical destruction'}\}$$

$$R(\{\textit{building}\}, \textit{demolish}) = \{\textit{'physical destruction'}\}$$

$$R(\{\textit{building}\}, \textit{ruin}) = \{\textit{'physical destruction'}\}$$

The first line here indicates that in the zero context ($c = \emptyset$) *destroy* is polysemous and it does not form one lexical domain. To give some examples consider the following three senses: to express the idea of defeating someone, as in *The home team demolished the visitors*; to describe physical destruction, as in *The fire destroyed the house*; and to indicate the termination of something abstract, as in *That argument destroyed their friendship*. Adding one lexical item (e.g. *building* as the c-element) to its context as a collocate reduces its polysemy and establishes a specific sense. Verbs such as *demolish* and *ruin* are used also in this sense forming a shared lexical domain with *destroy* when they co-occur with the same collocates in a specific syntactic and thematic context. Here, only one contextual element ($k=1$) is sufficient to establish individual senses of these verbs.

A corpus linguistic identification of near synonyms

Nothing in the present discussion hinges on the specific tools from distributional semantics or on the choice of *destroy* as the starting verb, which serves only as a convenient example; the same procedure could be applied to any other verb and lexical field. For the sake of convenience I use in the present study the Sketch Engine tools (Rychlý & Kilgarriff, 2007)

and more specifically the functionalities Thesaurus and Word Sketch Difference. Thesaurus is designed to identify sets of words with similar meaning based on their shared contexts (Rychlý & Kilgariff, 2007). One drawback, however, is that the algorithm underlying the Thesaurus tool has not been subjected to rigorous critical comparative analysis as is the case with other tools commonly used in distributional semantics. Nevertheless, some previous studies show that it does produce reliable results (e.g. Zakharov, 2018; Yuting, 2020; Hu & Yang, 2015). Similar to other tools based on the distribution hypothesis, Thesaurus generates lists of terms with semantic similarity based on collocational relations but it does not show contextual information. For example, among the top ten terms semantically most similar to *destroy* in the BNC are: *damage*, *abandon*, *affect*, *attack*, *create*, *lose*, *maintain*, *restore*, *save*, and *preserve*. However, *destroy* does not share the same collocational contexts with all of these terms. For example, while *destroy* and *lose* both collocate with *hope*, *maintain* does not. Conversely, *relationship* is a strong collocate of both *destroy* and *maintain*, but not of *lose*. This highlights the polysemic nature of *destroy* and it follows that these verbs cannot be polysemically reduced to the same lexical domain. In other words, they do not occur in $R(c, v)$ with the same c-element. These differences highlight the importance of analysing the specific contexts in which terms occur. Sketch Engine supports such analyses with the Word Sketch Difference tool, which compares the association strength of words based on the logDice metric (Rychlý, 2008). While Thesaurus lists similar terms, Word Sketch Difference allows direct comparison of two words at a time in terms of their common collocates; the c-element in $R(c, v)$. Table 1 illustrates how Word Sketch Difference operates to compare the collocational behaviour of two lexical items. In this specific example, the table displays whether nouns only occur with the verb *destroy*, the verb *damage*, or with both verbs. It also contains the information about frequencies and co-occurrence strengths of resulted collocations. The light-grey section contains only the collocates of *destroy*, the medium-grey section contains only the collocates of *damage*, and the dark-grey section contains nouns that collocate with both verbs. The Thesaurus algorithm draws on the values of collocation strength to identify semantically similar lexical items based on shared collocates. As mentioned above, the consideration of syntactic and thematic structures is crucial, because collocation relations alone are not sufficient for detailed lexical semantic analysis. Since the corpora in Sketch Engine have been tokenised, tagged with POS categories and syntactically parsed Word Sketch Difference allows the selection of specific syntactic contexts based on shallow grammatical relations. In this way, it is for example possible to extract the direct object nouns of *destroy* and *lose* (e.g. *destroy hope* and *lose hope*).

This functionality allows for a fine-grained comparison of how these verbs interact with their syntactic arguments and provides insights into their semantic behaviour.

lexical item	frequency with 'destroy'	frequency with 'damage'	collocation strength with 'destroy'	collocation strength with 'damage'
myth	10	0	6.4	—
dream	13	0	6.4	—
army	13	0	6.3	—
forest	25	4	7.6	5.5
life	44	11	6.5	4.6
bridge	19	4	7.1	5.4
site	24	6	6.8	5.1
crop	21	5	7.3	5.8
building	31	17	7.1	6.5
cell	22	12	6.9	6.4
tree	16	9	6.4	6
habitat	12	6	6.7	6.4
confidence	14	9	6.3	6.1
aircraft	22	16	7.2	7.2
morale	7	6	6	6.4
credibility	9	8	6.3	6.8
property	17	20	6.1	6.6
layer	14	14	6.7	7.3
fabric	7	7	5.8	6.4
vegetation	0	5	—	6.2
knee	0	7	—	6.3
engine	0	10	—	6.5
roof	0	8	—	6.5
tissue	0	7	—	6.5
skin	0	11	—	6.6

Table 1 Word Sketch Difference of the collocates associated with *destroy* and/or *damage*.

The data used in this analysis was collected in two stages using the BNC and Sketch Engine tools. The Thesaurus tool in Sketch Engine is used to generate a list of related verbs. The identification of near synonyms is operationalisation of the requirement that the verbs share the same collocate as the c-element in the reduction function and that it occurs either in the same syntactic position or is assigned the same thematic role or both. In particular, only the nominal collocates that occur in the direct object position or are assigned the thematic role of a Theme will be considered for the analysis. I will also introduce a rather generous cut-off point and include in the analysis those verbs that share at least one common collocate.

First, the verb *destroy* was used as a starting point and similar terms were identified using the Thesaurus tool. Collocates for these terms were then extracted using Word Sketch Difference. In subsequent iterations, the identified verbs served as search terms for the discovery of additional items, following the same procedure. For example, Thesaurus identified verbs such as *damage* and *ruin* as similar to *destroy*. These terms were then used as search terms to extend the data set. In order for a new term to be included in the study, its collocates had to meet certain requirements:

- The collocates must occur with at least two different lexical items.
- The collocates had to occur in a specific syntactic and semantic context.
- The resulting expression consisting of the verb and the nominal collocate must denote the event of destruction, even if the degree or mode of destruction may vary.

While the first two criteria are straightforward and follow on from the previous discussion, the third requires further explanation. Classifying identified verbs as ‘verbs of destruction’ is not an automatic output of corpus tools. As Stubbs notes “[w]hatever quantitative findings or statistics are produced by the computer, they must still be interpreted by the human analyst’ (Stubbs, 1996: 75). Similarly, McEnery and Hardie (2012) and McEnery and Brezina (2022) emphasise that corpus analysis is influenced by human decisions at various stages: when compiling corpora, designing annotation schemes, coding algorithms and interpreting results. Indeed, whenever data must be classified into semantic categories rather than simply ranked by statistical measures, human intervention is unavoidable. In this study, an interpretive step informed by the expertise of a linguist was applied to determine which of the candidate verbs genuinely denote events of destruction. This interpretation was guided by observations of distributional patterns and considering both syntactic criteria (e.g. agent–theme structure) and semantic considerations of event denoted.

The final list generated following the above requirements, includes the following verbs: *destroy*, *damage*, *ruin*, *demolish*, *harm*, *devastate*, *spoil*, *wreck*, *shatter* and *smash*. All these verbs denote the event in which an Agent (used in the subject position) performed an act of destruction that affected a Theme (used in the direct object position). In the present study, therefore, the nominal collocates in transitive argument structure have both the same syntactic function and are assigned the same thematic role. But, unlike other verbs *smash* and *shatter* can in addition to transitively also be used intransitively. The noun phrases *the vase* and *the*

glass are still Themes, even in these intransitive constructions. This is because they are the entities undergoing the change of state caused by the events described by the verbs.

Lexical domain of ‘verbs of destruction’: general distributional properties

As already mentioned the following list of verbs was identified using the BNC and the procedure described in Section 4: *destroy*, *damage*, *ruin*, *demolish*, *harm*, *devastate*, *spoil*, *wreck*, *shatter* and *smash*. In order to identify the distributional patterns of these verbs they were examined pairwise for all possible combinations and with 10 verbs it means that there are 45 unique pairwise combinations. For each pair, common collocates were identified and stored for further analysis using Word Sketch Difference. I will now first describe the general nature of the data obtained and in the next section propose metrics that capture behavioural patterns, which will be used to incorporate the probability aspect of language use into the formalisation described above. All results were obtained by writing R scripts operated on the data from the BNC.

The heatmap in Figure 1 visualises the association strength based on the logDice values for the collocations consisting of the destruction verbs and the shared nouns occurring in the direct object position and in the subject position in the case of *smash* and *shatter*. Rows represent individual nouns and columns represent verbs, with colour intensities indicating scaled association strengths. Darker shades indicate higher association strengths, while lighter shades represent weaker or negligible associations. The heatmap reveals patterns of similarity through clustering of verbs with comparable association strengths to individual nouns. For example, the cluster containing *destroy* and *damage* is due to their strong association with the nouns *building*, *bridge*, *area*, *career* and *economy*. Notice that the association strength does not necessarily indicate the similarity in the semantic content across individual collocates. In the present example the first three nouns indicate spatial entities the latter two indicate abstract notions and therefore the resulting collocations will have slightly different semantic content (see Trklja, 2020 for more details). By calculating the average association strength for each verb within these clusters, it is possible to rank the verbs according to the shared collocates.

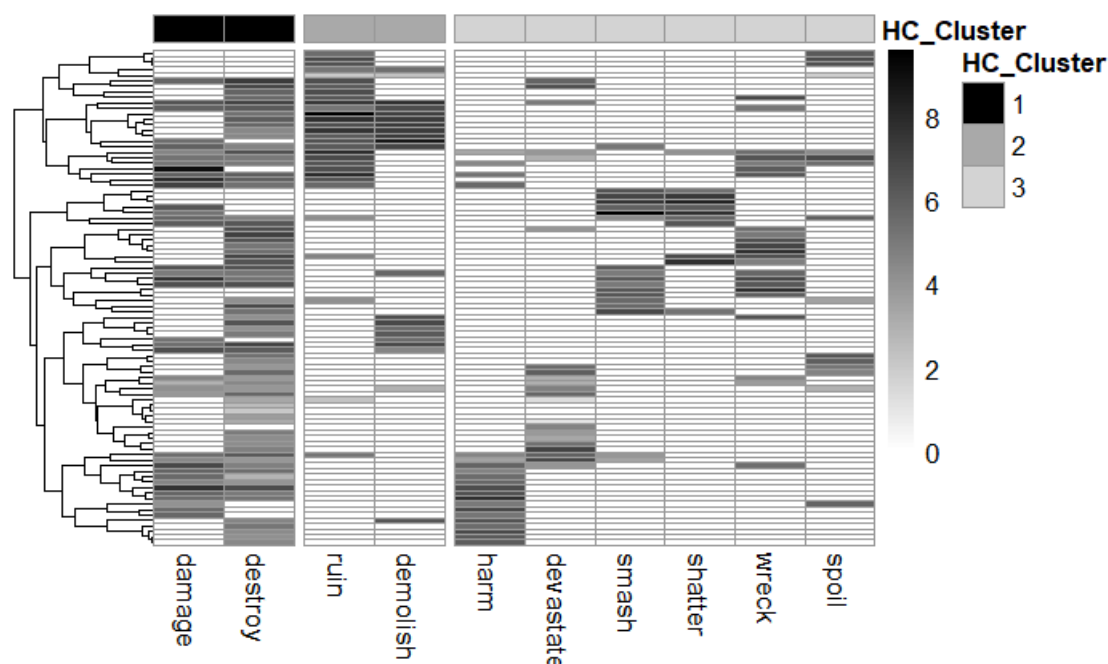


Figure 1 Clusters of verbs based on the shared nominal collocates.

At the next stage I explored the clustering of verbs by exploring their behaviour in terms of skewness and kurtosis. These measures can usefully capture both the central tendencies and the variability and extremes of verb distributions in the present data. Skewness measures the asymmetry of a distribution, indicating whether data points are spread more to one side of the mean than the other. Positive skewness reflects a long right tail (e.g. extreme high values), while negative skewness suggests a long left tail (e.g. extreme low values). In contrast, kurtosis quantifies the ‘tailedness’ of a distribution, highlighting the presence and extremity of outliers. Roughly speaking, a normal distribution has moderate kurtosis, while high kurtosis indicates heavy tails with more extreme values, and low kurtosis indicates lighter tails. In the present analysis using hierarchical clustering, the verbs with similar skewness and kurtosis values were grouped into distinct clusters. Three specific clusters emerged in the present data: verbs with (1) low skewness and low kurtosis, suggesting symmetric distributions with minimal outliers; (2) high skewness and low kurtosis, suggesting asymmetric distributions dominated by specialised contexts; and (3) high skewness and high kurtosis, characterised by heavy-tailed distributions with rare but significant occurrences. The scatterplot in Figure 2 displays these clusters with each point representing a verb. The x-axis and y-axis reflect the skewness and kurtosis of each verb’s distribution, respectively.

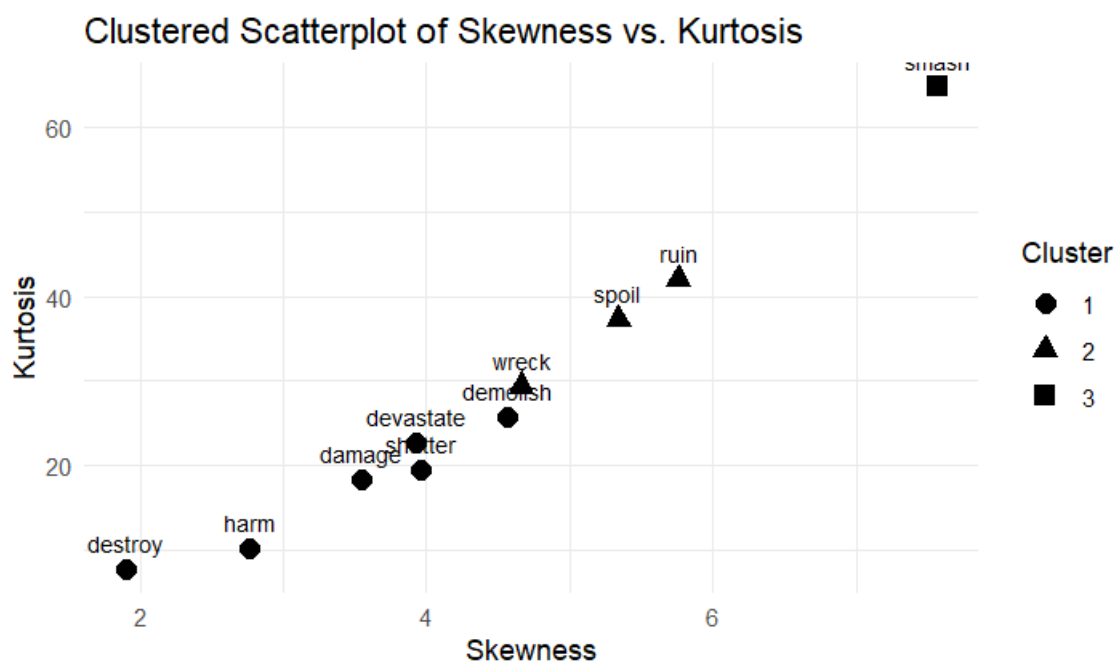


Figure 2 Three clusters of verbs of destruction based on skewness vs kurtosis.

Distribution metrics and usage patterns of ‘verbs of destruction’

The above observations indicate that the verbs in the current lexical domain differ in the number of collocates and the association strength. In order to further analyse the distributional behaviour of these verbs, I propose three metrics: Total Strength, Breadth, and Average Strength, which collectively inform a composite what I referred to as Typicality Index. Total Strength reflects the sum of a verb’s association strengths across all nominal collocates. Breadth measures the diversity of a verb’s usage defined as the number of nouns with which it is associated indicating how widely the verb is used. Finally, Average Strength focuses on the intensity of the associations by calculating the mean strength of a verb’s associations with its nominal collocate. These metrics were normalized to allow comparison and combined into a weighted Typicality Index to identify verbs most representative of the class. The following formula was used here.

$$\text{Typicality Index} = w1 \cdot \text{Norm Total Strength} + w2 \cdot \text{Norm Breadth} + w3 \cdot \text{Norm Average Strength}$$

All metrics¹ are treated as equally important in order to avoid subjective bias in assigning weights. Besides, equal weights let the data’s inherent relationships determine the results

¹ Total Strength and Average Strength are based on the concept of association or collocational strength. These metrics are based on the concept of collocation, extending it to allow examination of the distribution of

reducing the risk of self-fulfilling conclusions. Any observed trends (i.e., the importance of a specific metric in predicting Typicality Index) will therefore not be artifacts of the weighting scheme. The results of the statistical analyses are displayed in Figures 3 to 5.

The results reveal that one verb (*destroy*) emerges as the most typical lexical item within the lexical domain examined. The typicality of this verb cannot be explained by its occurrence in the zero context ($c=\emptyset$), because like all other verbs it is examined in the context of its collocates. This result is rather due to its distributional properties which are clearly reflected in the metrics. This verb has the highest scores for the Total Strength (368) and Breadth (71) which underscore its broad usage and strong overall associations with the shared nominal collocates. Although its Average Strength (5.19) is in this case moderate relative to some other verbs, its combination of wide usage and substantial impact makes it the most representative verb here. Similarly, *damage* and *ruin* rank highly, with significant Total Strengths (223 and 181, respectively) and Breadth (39 and 29), reflecting their broad but slightly less dominant usage compared to *destroy*. In contrast, verbs like *shatter* and *demolish* occur with higher Average Strength (6.41 and 6.26, respectively), indicating their intense but more context-specific associations with collocates, as evidenced by their narrower Breadth (11 and 21). These findings point out the dual importance of both breadth and intensity in determining a verb's typicality, because verbs like *destroy* balance broad applicability across different collocates, while specialized verbs such as *shatter* demonstrate strength in niche contexts but lack the diversity to be considered broadly typical.

association strengths for individual collocations within the same lexical domain. Breadth does not correspond to any established concept in corpus linguistics, nor does the Typicality Index, which combines the results of the three metrics to identify the most representative verbs of a given class. Colligation is related to any metrics but it is related to the discussion of argument structure, which is one of the criteria used to identify shared collocates. However, colligation is probabilistic (i.e. the likelihood of a lexical item occurring with a specific syntactic category), whereas argument structure is categorical and lexically specific, defining the set of syntactic and semantic roles that a verb can fulfil. Semantic preference and semantic prosody are not directly related to the proposed metrics.

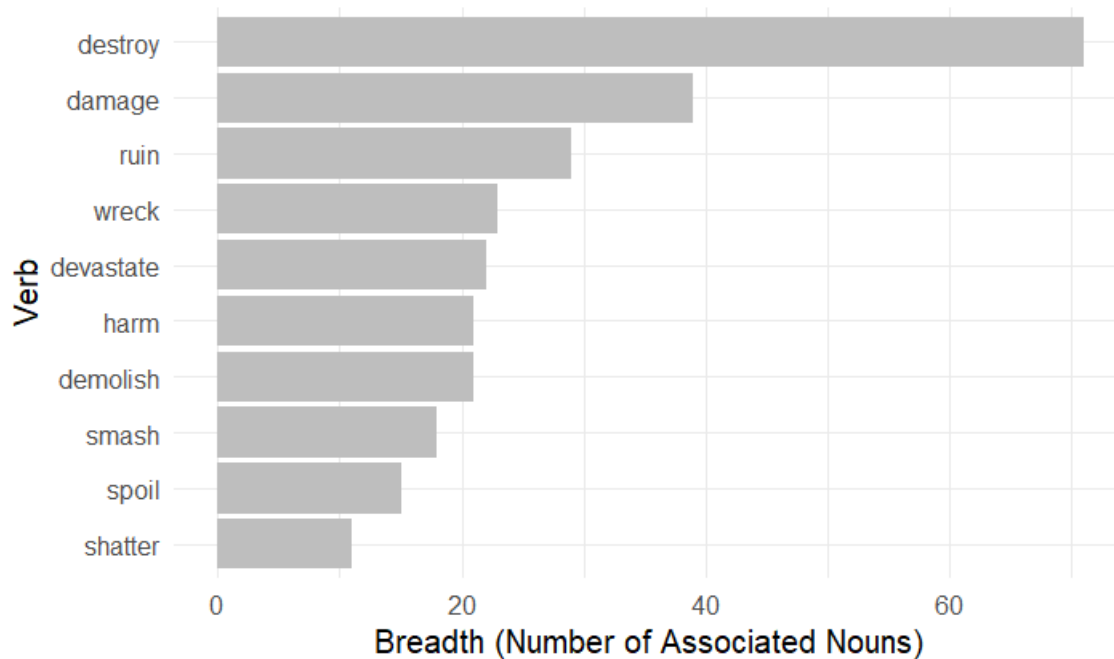


Figure 3 The size of the sets of nominal collocates associated with the ‘verbs of destruction’.

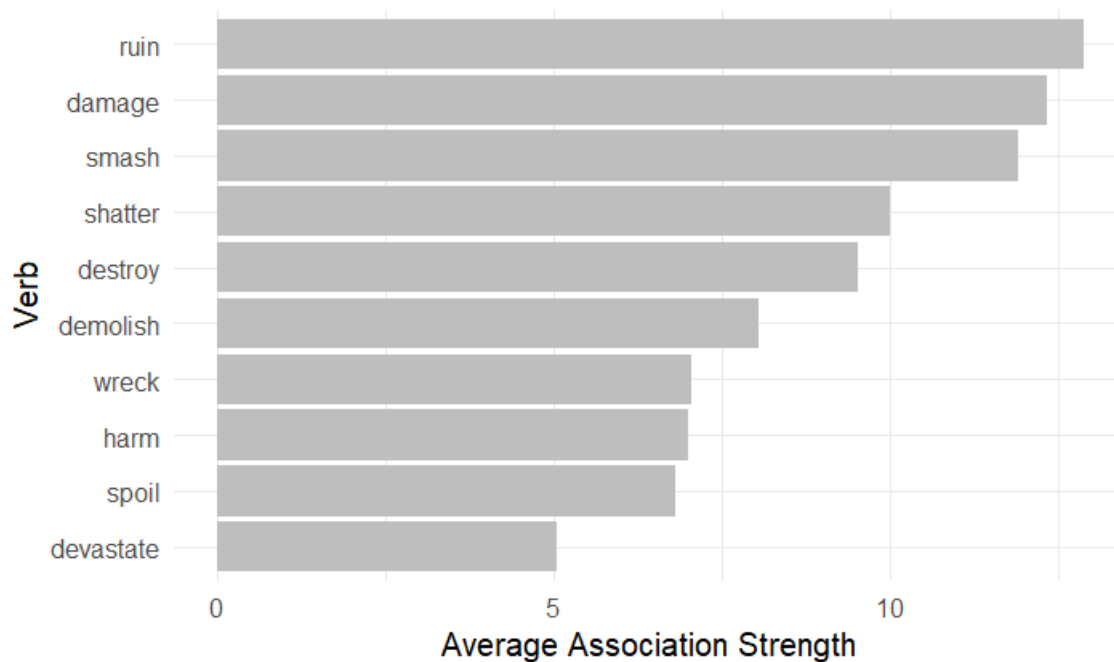


Figure 4 The mean strength of a verb’s associations with its nominal collocate.

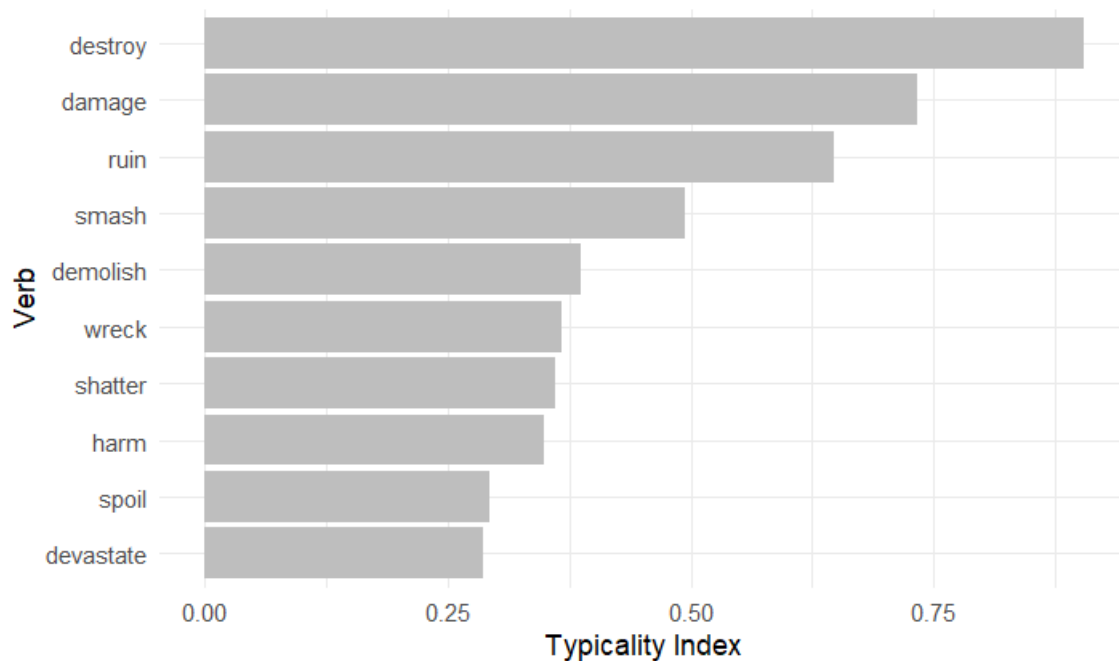


Figure 5 Typicality index of ‘verbs of destruction’.

The above results can further be examined and interpreted by using pairwise correlation matrix as displayed in Figure 6. In particular, these results indicate the relationships between the individual metrics. The graph confirms a strong linear relationship between Total Strength and Breadth ($r=0.987$), indicating that verbs with broader associations across nouns tend to accumulate greater overall impact. This is consistent with the observation that verbs such as *destroy* and *damage* excel in both metrics which reflects their typicality in the lexical domain. Additionally, the Typicality Index correlates almost perfectly with Total Strength ($r=0.995$) and strongly with Breadth ($r=0.972$), confirming that typical verbs are characterised by both high impact and diversity of collocates. On the other hand, Average Strength exhibits much weaker correlations with both Breadth ($r=0.161$) and Total Strength ($r=0.298$), suggesting that while some verbs (e.g., *shatter* and *demolish*) have strong associations in specific contexts their narrow usage limits their overall typicality. This weak link between Average Strength and the other metrics highlights its role in identifying specialized verbs which are significant in only certain specific contexts but are less representative of the broader class of verbs. Importantly, these relationships stress the multifaceted nature of typicality, where impact (Total strength), diversity (Breadth), and intensity (Average strength) play distinct but interrelated roles.

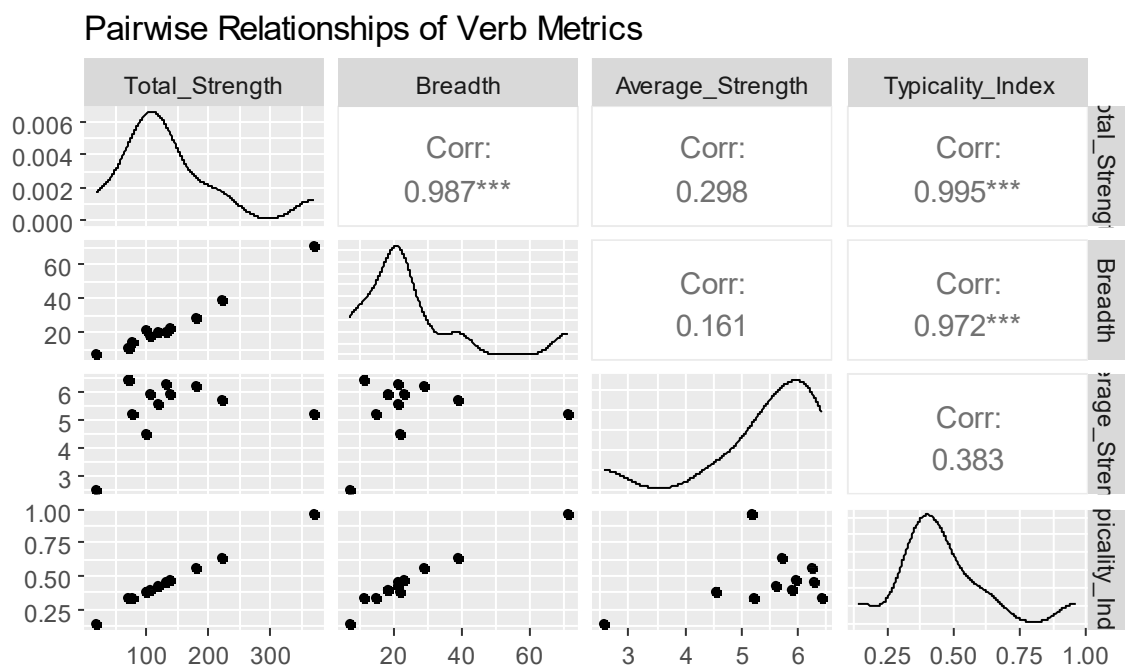


Figure 6 Correlation relations between ‘verbs of destruction’.

The analysis suggests that Breadth is a key predictor of a verb’s Typicality Index, as verbs with broader associations across nouns tend to achieve higher typicality scores. ($\beta = 0.00944$; 95% *CI* [0.0079 – 0.0110]; $R^2 = 0.961$; $F(1,8) = 197.6$; $p < .001$). Each additional associated noun increases the Typicality Index by about 0.0094. This is supported by the strong correlations between Breadth and both Total Strength ($r=0.987$) and the Typicality Index ($r=0.972$), indicating that diversity of usage heavily influences a verb’s representativeness. Verbs such as *destroy* and *damage*, which are associated with many nouns, exemplify this relationship, achieving high typicality due to their widespread and impactful usage. Conversely, verbs with narrow Breadth, like *shatter* and *demolish* often exhibit high Average Strength but fail to achieve high typicality, as their specialized usage limits their overall representativeness. Average Strength, while reflecting the intensity of associations, is only weakly correlated with Breadth ($r=0.161$) and Total Strength ($r=0.298$), making it less predictive of typicality. These findings suggest that in similar datasets, verbs with greater Breadth and proportional Total Strength will likely dominate typicality rankings, while verbs with high Average Strength will be more specialised and less representative of the broader verb class. In order to make sure that the dominance of Breadth in predicting Typicality Index is not merely an artifact of weighting but a reflection of the data’s structure I tested alternative weightings, to evaluate whether the rankings remain stable or change dramatically. Regardless of the weighting Breadth consistently predicts high typicality even if Average strength is

prioritized in relation to other two factors. Therefore, the findings are likely driven by the data and Breadth plays a pivotal role in typicality because it captures diversity of usage, while Average strength contributes less due to its weak relationship with Breadth and Total Strength.

Besides, the verbs with high Total Strength contribute more to the Typicality Index which indicates the cumulative effect which emerges due to their Breadth values (broad usage). This cumulative effect can be explained by power-law distributions. In any system governed by power-law distributions, a small number of elements have disproportionately high values while most elements contribute minimally. This can explain why Total Strength correlates so strongly with Typicality in our data: the contributions of high-strength verbs far outweigh the cumulative impact of weaker verbs. This is a manifestation of the Matthew effect, epitomised by the principle ‘the rich get richer and the poor get poorer’. This principle reflects a self-reinforcing dynamic where entities that already have advantages continue to accumulate them while those with fewer advantages struggle to catch up. In the present data, verbs with higher Total Strength dominate the Typicality Index because their broad and strong associations amplify their impact, leaving verbs with fewer or weaker associations with limited influence. When a verb already has high Breadth (broad usage) and Total Strength (cumulative impact), it attracts more associations over time, reinforcing its dominance. Broadly used verbs are likely to be used even more, amplifying their impact. The strong correlation between Total Strength and Typicality Index mirrors, of course, the Zipfian behaviour where high-frequency elements disproportionately contribute to the overall structure.

In terms of cognition, the observed dominance of a few high-typicality verbs accords with the theories of frequency effects in lexical access according to which established usages are more likely to be quickly recalled from memory, while less established expressions require more cognitive effort to access (Forster, 1976; Heitmeier et al, 2024). This is because words or phrases encountered more frequently are stored with stronger mental representations and are thus more readily activated during language processing. Similarly, the exemplar theory of memory (e.g. Goldinger, 1998; Kleinschmidt and Jaeger, 2015) emphasizes that frequently encountered linguistic items are encoded with multiple exemplars, strengthening their mental representation over time. Furthermore, studies on usage frequency (e.g., Durrant & Schmitt, 2010) demonstrate that frequent co-occurrences between verbs and nouns are more likely to be stored as fixed chunks or associations in memory facilitating faster recall and usage. Therefore, it can be assumed that verbs which have higher Breadth and Total Strength likely reflect entrenched patterns of usage that are cognitively reinforced over time. This phenomenon has

been extensively observed in cognitive linguistics (Langacker, 1987; Divjak and Caldwell-Harris, 2019)), which emphasize that linguistic knowledge emerges from patterns of use and is shaped by frequency and context. In particular, Bybee (2006, 2010) argues that high-frequency words and constructions are stored as stronger cognitive units due to repeated exposure, making them more accessible and resistant to change.

Concluding remarks

The present study has achieved two main goals. First, it has proposed a formalised account of polysemy reduction that allows for a formal characterisation of near-synonyms and lexical domains. Second, it has introduced specific metrics for studying the patterns of distributional properties of near-synonymous expressions from the same lexical domain.

The study began by formalising polysemy in terms of degrees of contextual specification, illustrating how contextual elements incrementally disambiguate meanings. This approach provides a principled method for analysing synonymy and its interaction with polysemy, offering a deeper understanding of meaning variation within lexical domains. By incorporating syntactic arguments and thematic roles alongside collocational relations, the study formulated clear criteria for a systematic exploration of lexical meaning and for identifying near-synonymous expressions. This dual focus on distributional and syntactic properties enhances the explanatory power of the proposed methodology. In previous corpus linguistics and distributional semantics studies of near synonyms, insufficient attention has been paid to the formalisation of lexical semantic relations within a lexical item. While corpus linguistics relies on the observation of patterns in data, it typically neglects the importance of formalised principles and concepts. However, as we know from other disciplines, formalisation ensures clarity in the definition of concepts and provides a structured basis for empirical analysis. Of course, formalisation is never an end in itself, but a means to strengthen the analytical power of an empirical approach, allowing for more consistent, replicable and theoretically grounded findings. In distributional semantics, formalism is a familiar feature, but it is primarily applied to the formalisation of metrics rather than to theoretical concepts such as context, meaning, and polysemy, as is the case here. The present formalisation allows exploration of various empirical issues using corpora. First, empirical analysis can explore whether there are any tendencies with respect to how much context (*c*) is typically required to reduce polysemy of lexical items in specific language, genre or register and to identify lexical domains. Second, by analysing a range of words, including both single words and multiword

expressions, we can compare the degree of polysemy and the stabilisation points $k(w)$. For example, do highly frequent verbs (such as *run*, *make*, *get*) require more context to resolve ambiguity than less frequent or more specialised verbs? Third, one can examine the patterns of disambiguation for individual words by comparing whether different senses of, say, a polysemous verb are established with the same cardinality. One of the key questions relevant to the study of lexical domains is how verbs within the same lexical domain behave in terms of stabilisation points and ambiguity resolution. For example, are verbs from the same lexical domain disambiguated with the same cardinality? Due to space constraints, it was not possible for the present paper to do justice to any of these empirical investigations.

The metrics proposed - Total Strength, Breadth, Average Strength and Typicality Index - allowed for a systematic examination of the relationships between synonymous lexical units. In particular, the Typicality Index provided a means of measuring the centrality of a lexical item within the domain. In addition, analysing the contribution of each metric facilitated an exploration of their predictive power in capturing distributional relationships. While the study focused on a single lexical domain and dataset, the metrics are broadly applicable to other semantic domains, languages and corpora. The metrics made it possible to investigate different aspects of the verbs' behaviour within the lexical domain, such as the cumulative effect of association strength, generality vs. specificity, and broad applicability vs. niche use. The results are also indicative of the structure of lexical domain. Verbs with high Typicality Index can be identified as core members of the lexical domain, while those with low Breadth but high Average Strength as more peripheral. The present case study demonstrated the specific relationships between individual metrics and the strong predictive power of Breadth and Total Strength with respect to the Typicality Index. Larger scale studies are needed to show whether all the mentioned properties are specific of the present data or represent a general tendency.

One limitation of the present study is its focus on synonymous terms. Other types of semantic relation, such as antonymy, hypernymy and meronymy, were not included, primarily due to space limitations and the fact that addressing them would have taken the study in different conceptual directions (e.g. lexical contrast or taxonomic structures). Nevertheless, I believe that the formal approach and metrics proposed in this paper could be extended to these other types of semantic relation with minimal modification. However, only further theoretical exploration and empirical investigation can determine whether this is true.

Besides, while the present paper illustrated the proposed approach through a single case study, a large-scale investigation is needed to determine whether the observed patterns reflect properties specific to the examined lexical domain or indicate broader, potentially universal tendencies in lexical organisation. For example, it remains unclear whether all lexical domains contain certain lexical items with the highest Typicality Index due to systematic interactions between association strength and the number of collocates that a node word can attract, or if this tendency depends on additional factors. A broader empirical study is also necessary to explore how much contextual information is typically required to reduce polysemy, and whether regularities can be observed in the stabilisation points $k(v)$, including variation across part of speech, register, or corpus type. Such large-scale analyses would provide a more robust empirical basis on which to evaluate the generalisability and explanatory power of the proposed framework.

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